Retail Demand Forecasting & Inventory Optimization

1. Project Objective

The goal of this project is to help a retail organization forecast product demand, optimize inventory levels, and reduce stockouts by:

- Predicting monthly sales across stores and SKUs
- Calculating safety stock and reorder points
- Identifying high-risk stock items
- Delivering actionable insights for inventory planning and merchandise management

2. Dataset Description:

Dataset Source: Kaggle - Store Item Demand Forecasting (train.csv)

Dataset Characteristics:

Column	Description
date	Daily sales date
store	Store ID (1–10)
item	Product ID (1–50)
sales	Units sold

Dataset Size: ~913,000 rows covering 5 years of daily sales data

Key Features Used:

- Historical sales (sales)
- Store and item identifiers (store, item)
- Derived time features (year, month, year_month)

3. Data Preparation:

Steps Taken:

1. Converted date column to datetime format
2. Created time features: year, month, day, week
3. Aggregated data to monthly level per store and item for forecasting
4. Checked for missing values and cleaned data

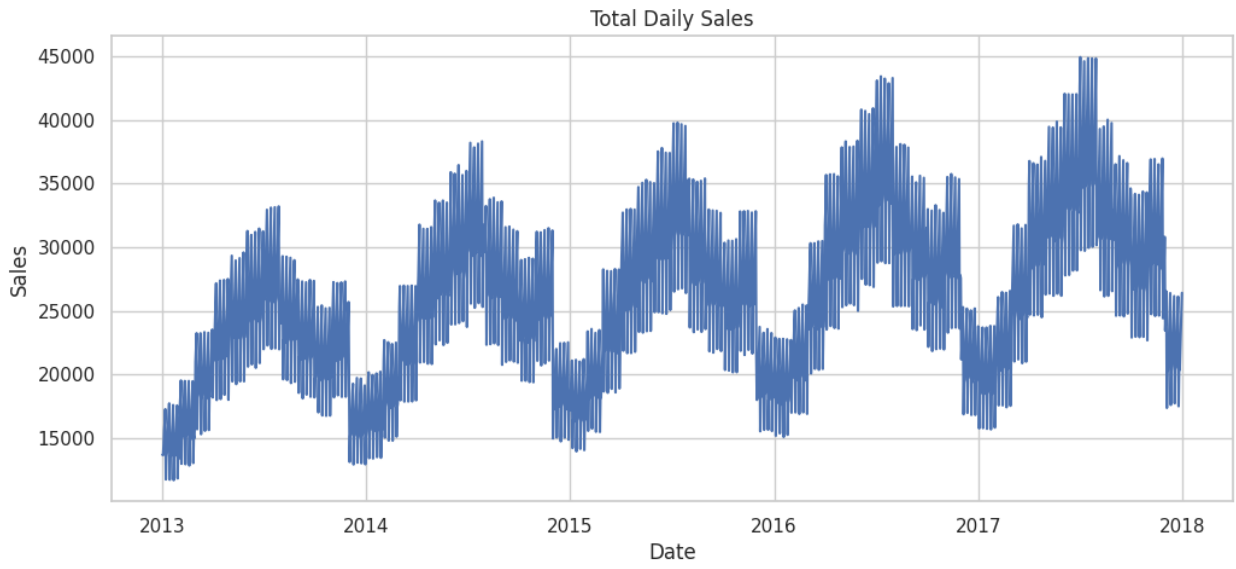
4. Exploratory Data Analysis (EDA):

Key Observations:

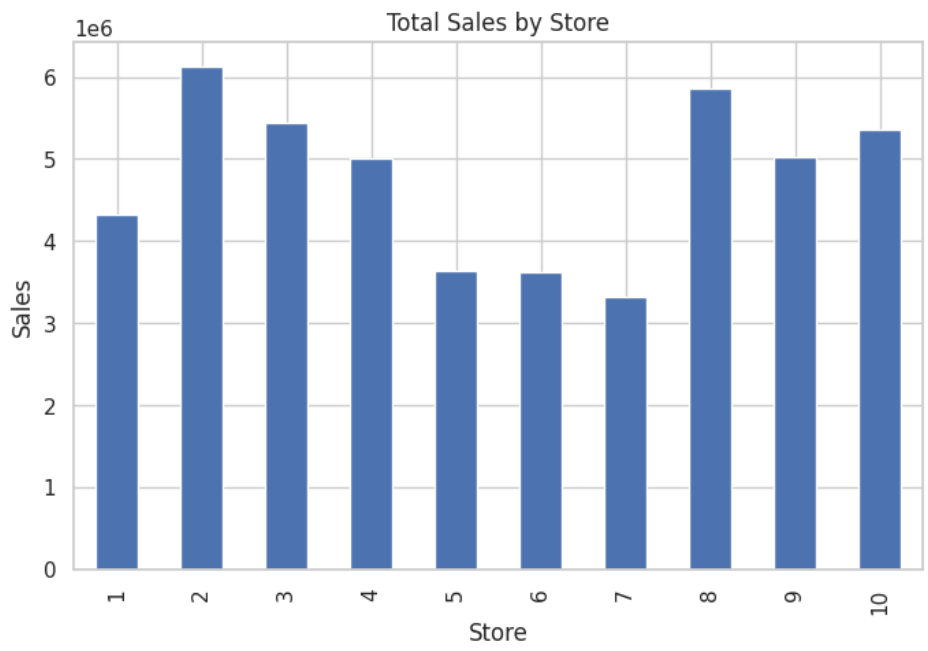
- Overall sales trend: Stable upward trend with seasonal spikes
- Store-level sales: Certain stores generate higher sales consistently
- SKU-level sales: Top-selling items dominate total sales volume
- Monthly seasonality: Peaks in demand occur during specific months

Sample Visuals (Python / Matplotlib):

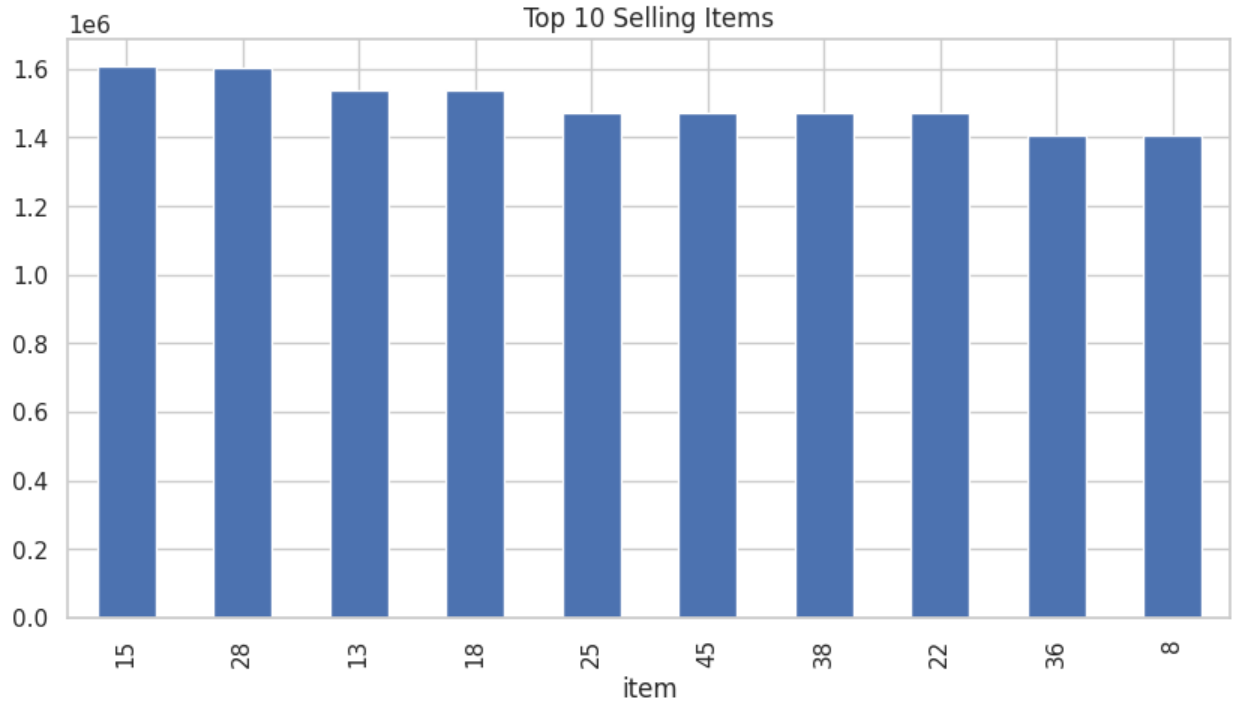
1. Total Daily Sales Trend



2. Total Sales by Store



3. Top 10 Selling Products



5. Forecasting Model:

Methodology:

- Aggregated monthly demand per store and SKU
- Applied 3-month moving average to forecast future demand
- Evaluated forecast using Mean Absolute Error (MAE)

Forecast Output Example:

year_month	store	item	sales	forecast
2013-01	1	1	120	118
2013-01	1	2	98	95

Insight: Forecast closely follows historical demand patterns and captures seasonality trends.

6. Inventory Optimization:

Metrics Calculated:

Metric	Formula	Purpose
Average Demand	Mean of monthly sales	Base demand level
Demand Variability	Std deviation of monthly sales	Measures fluctuations
Safety Stock	Z × Std × √Lead Time	Buffer against variability
Reorder Point	Avg Demand × Lead Time + Safety Stock	Trigger replenishment
Stockout Risk	Current Inventory < Reorder Point	Identifies high-risk SKUs

Assumptions:

- Service Level = 95% → Z = 1.65
- Lead Time = 7 days
- Simulated current inventory: 100–500 units

Sample Output Table:

store	item	avg_demand	std_demand	safety_stock	reorder_point	current_inventory	stockout_risk
1	1	100	20	73	803	450	High Risk
2	5	95	15	63	698	350	Safe

7. Key Insights:

1. High-demand SKUs require priority in replenishment
2. Stores 5 and 8 consistently have higher sales → allocate inventory accordingly
3. Stockout risk detected in several SKUs → adjust safety stock levels
4. Seasonality observed → plan inventory for peak months

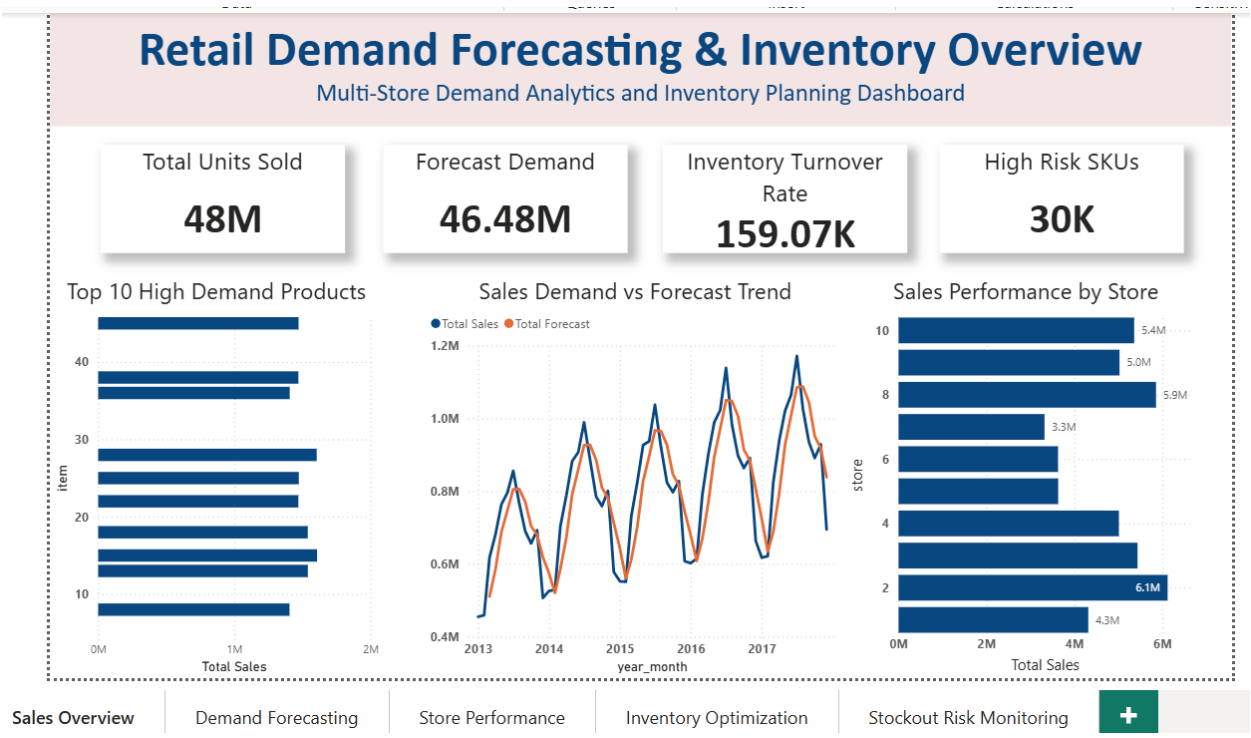
8. Conclusion:

This project demonstrates:

- End-to-end retail demand analytics from historical sales to actionable insights
- Inventory optimization using forecast-driven metrics

Impact: Enables businesses to reduce stockouts, improve inventory planning, and make data-driven decisions.

9. Next Steps / Recommendations:

1. Integrate promotions, holidays, and seasonal events to improve forecast accuracy
2. Test advanced forecasting models (ARIMA, Prophet, LSTM) for better precision
3. Incorporate real inventory and supplier lead time data for dynamic reorder points
4. Extend analysis to category-level and multi-store optimization

10. Power BI Dashboard: (Link)